

Q1 - 24 January - Shift 1

If $\vec{E}$ and $\vec{K}$ represent electric field and propagation vectors of the EM waves in vacuum, then magnetic field vector is given by: (ω - angular frequency) :

- (1) $\frac{1}{\omega}(\vec{K} \times \vec{E})$ (2) $\omega(\vec{E} \times \vec{K})$
(3) $\omega(\vec{K} \times \vec{E})$ (4) $\vec{K} \times \vec{E}$

Space for your notes:

Q2 - 24 January - Shift 2

The electric field and magnetic field components of an electromagnetic wave going through vacuum is described by

$$E_x = E_0 \sin(kz - \omega t)$$

$$B_y = B_0 \sin(kz - \omega t)$$

Then the correct relation between E_0 and B_0 is given by

- (1) $kE_0 = \omega B_0$
(2) $E_0 B_0 = \omega k$
(3) $\omega E_0 = kB_0$
(4) $E_0 = kB_0$

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Q3 - 25 January - Shift 1

All electromagnetic wave is transporting energy in the negative z direction. At a certain point and certain time the direction of electric field of the wave is along positive y direction. What will be the direction of the magnetic field of the wave at that point and instant?

- (1) Positive direction of x
- (2) Positive direction of z
- (3) Negative direction of x
- (4) Negative direction of y

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Q4 - 29 January - Shift 1

Which of the following are true?

- A. Speed of light in vacuum is dependent on the direction of propagation.
- B. Speed of light in a medium is independent of the wavelength of light.
- C. The speed of light is independent of the motion of the source.
- D. The speed of light in a medium is independent of intensity.

Space for your notes:

Choose the correct answer from the option given below :

- (1) A and C only
- (2) B and D only
- (3) B and C only
- (4) C and D only

Q5 - 29 January - Shift 2

Given below are two statements:

Statement I : Electromagnetic waves are not deflected by electric and magnetic field.

Statement II : The amplitude of electric field and the magnetic field in electromagnetic waves are

related to each other as $E_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} B_0$

In the light of the above statements, choose the correct answer from the options given below:

- (1) Statement I is true but statement II is false
- (2) Both Statement I and Statement II are true
- (3) Statement I is false but statement II is true
- (4) Both Statement I and Statement II are false

Space for your notes:

Q6 - 31 January - Shift 2

Match List-I with List-II.

	List-I		List-II
A.	Microwaves	I	Physiotherapy
B	UV rays	II	Treatment of cancer
C	Infra-red rays	III	Lasik eye surgery
D	X-rays	IV	Aircraft navigation

Space for your notes:

Choose the correct answer from the option given below:

- (1) A-II, B-IV, C-III, D-I
- (2) A-IV, B-I, C-II, D-III
- (3) A-IV, B-III, C-I, D-II
- (4) A-III, B-II, C-I, D-IV

Q7 - 01 February - Shift 1

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Match the List-I with List-II:

	List I		List II
A.	Microwaves	I.	Radio active decay of the nucleus
B.	Gamma rays	II.	Rapid acceleration and deceleration of electron in aerials
C.	Radio waves	III.	Inner shell electrons
D.	X-rays	IV.	Klystron valve

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Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-III, D-IV
- (2) A-IV, B-I, C-II, D-III
- (3) A-I, B-III, C-IV, D-II
- (4) A-IV, B-III, C-II, D-I

Q8 - 01 February - Shift 2

The ratio of average electric energy density and total average energy density of electromagnetic wave is:

- (1) 2
- (2) 1
- (3) 3
- (4) $\frac{1}{2}$

Space for your notes:

Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (1)

Q2 (1)

Q3 (1)

Q4 (4)

Q5 (1)

Q6 (3)

Q7 (2)

Q8 (4)

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Q1 (1)

Magnetic field vector will be in the direction of

$$\hat{K} \times \hat{E}$$

$$\text{magnitude of } B = \frac{E}{C} = \frac{K}{\omega} E$$

$$\text{Or } \vec{B} = \frac{1}{\omega} (\vec{K} \times \vec{E})$$

Q2 (1)

$$C = \frac{\omega}{k} = \frac{E_0}{B_0}$$

Q3 (1)

As, poynting vector

$$\vec{S} = \vec{E} \times \vec{H}$$

Given energy transport = negative z direction

Electric field = positive y direction

$$(-\hat{k}) = (+\hat{j}) \times [\hat{i}]$$

Hence according to vector cross product magnetic field should be positive x direction.

Q4 (4)

Speed of light does not depend on the motion of source as well as intensity.

Q5 (1)

Statement-I is correct as EMW are neutral

Statement-II is wrong.

$$E_0 = \sqrt{\frac{1}{\mu_0 \epsilon_0}} B_0$$

Q6 (3)

No Solution

Q7 (2)

Based on theory

Q8 (4)

$$\langle u_E \rangle = \langle u_B \rangle = \frac{1}{2} \langle u_{\text{total}} \rangle$$

$$\text{So } \frac{\langle u_E \rangle}{\langle u_{\text{total}} \rangle} = \frac{1}{2}$$